

Boundary scan

Boundary scan is the unique solution to many test requirements and provides information about the board without accessing the complete board. It is ideal for testing those complex boards that could not be tested otherwise due to lack of test access.

Boundary scan systems have little hardware and their significant part is software. The well-established test technique boundary scan JTAG IEEE Std. 1149.x requires test program to be generated before it can be used. Being an integral part of a boundary scan system, test program generator uses the net list of the unit under test and the BSDL files of the boundary scan components contained within the circuit to create test patterns for the test. The test program generator also creates test vectors that enable the system to detect faults on the nodes and non-boundary scan components that are surrounded by boundary scan devices.

Functional automated test

Digital boards are well suited for testing with functional automated test (FATE) systems where much of the program can be generated automatically by entering the circuit data into the tester, which builds up its own picture of the circuit and then, with knowledge of the pin connections, it can build up a test program for the board. The simulations can reveal design problems such as race states or even the non-required circuitry. Unfortunately, the program generation usually needs a lot of finishing, which is a time-consuming process. Also, the analogue areas often require analogue-measuring instruments to be used and need to be programmed manually. This manual programming can be very expensive to implement.

Although, FATE systems can be very fast at finding functional faults with a board, they are not always so fast in finding the problem area. In most cases they are unable to locate a problem because of lack of view of the internal areas of the board. A guided probe can be connected to the

TABLE III ICT Testing Solutions for PCBs		
Vendor	ICT Solution	Key Features
Teradyne	Test Station LH	Low cost of ownership Most accurate in-circuit test pin ever developed: UltraPin Driver/Sensor Real-time backdrive current measurement reports
Agilent Technologies	i1000D	Digital-to-analogue and analogue-to-digital test strategies. Latest VTEP v2.0 vectorless test suite, which comprises the award winning iVTEP and incorporates the new network parameter measurement (NPM) technology targeted at defect detection on power and ground pins for connectors. Wired/wireless options for fixtures
Aeroflex	4200 Series	Single box solution: Contains all power supplies and controller Full Analogue and digital in-circuit GPIO and CAN bus options Boundary Scan (JTAG IEEE1149.1) test integration
Spea	3030 Inline	Multicore technology for true parallel tests on several boards 4x throughput as compared to other bed of nails systems The True Per Pin Architecture makes all the analogue and digital resources always available on each system channel

TABLE IV FATE Testing Solutions for PCBs		
Vendor	FATE Solution	Key Features
Agilent Technologies	TS-8900	Test Exec SL 7.0 & TS-5000 system software with over 400 built-in test libraries 32-channel high-voltage acquisition modules, large voltage-current range V/I source and a high-voltage 8/16 channel DAC enabling parallel test methodologies and built-in 400 application-tuned libraries for faster deployment
Acculogic, Inc.	Tracer 2000	High-speed analogue and digital functional test—VXI, PXI and GPIB-based instrumentation Boundary Scan (JTAG IEEE1149.1) test integration LabWindows CVI, Labview, Visual C/C++ , etc
Aeroflex	5800 Series	Teststand, Labview, C# and VB.NET or any other platform that is .NET compliant. Analogue-ICT test integration Full analogue and digital functional test

tester that can be manually applied to different points on the circuit under program control to check the points on the board that are not accessible via the bed of nails. Again, in the case of analogue areas, the routines required for fault finding using guided probe need to be programmed manually, which can be particularly time consuming.

Design for testability

Boundary scan—and any other test technology—requires design rules that must be considered. If such design rules are disregarded, the achievable test depth might be considerably affected, or in extreme cases, completely

lost. “Nothing is ‘sadder’ than a board that cannot be tested because of one missing interconnection. But there’s no need to worry about possibly ‘many’ design rules. Convenient software provides support for rule compliance,” adds Harish. Moreover, it once again demonstrates that it makes sense to start with test generation at a very early stage of product design. Once the layout is finalised, things are relatively hard to change. ●

Pankaj V. is a technical journalist at EFY, Gurgaon, while Dilin Anand is senior technical correspondent at EFY Bengaluru. You can read the full story on www.electronicsonline.com in the Test and Measurement section